CYBER SECURITY, CII AND CYBERCRIME

CIA IMPACT MAP → CYBER-RISK CHAIN → INCIDENT-CRIME-OPERATION MATRIX → ATTRIBUTION CONFIDENCE LADDER → IT ACT CII LADDER → CYBERCRIME GOVERNANCE CHAIN

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◆ SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

00 →

STAGE 00 CIA IMPACT MAP

CIA IMPACT MAP · UNAUTHORISED DISCLOSURE · UNAUTHORISED ALTERATION · TRUST LOSS · DENIAL OF SERVICE · IDENTIFY THE PROPERTY BEFORE THE REMEDY

CONFIDENTIALITY → unauthorised disclosure

INTEGRITY → unauthorised alteration / trust loss → **AVAILABILITY → disruption / denial of service** → **RULE → identify the property before the remedy**

THREAT / GOVERNANCE CONCEPT
- CONFIDENTIALITY → unauthorised disclosure
- INTEGRITY → unauthorised alteration / trust loss

INSTITUTIONAL RESPONSE
- AVAILABILITY → disruption / denial of service
- RULE → identify the property before the remedy

LEGAL / EVIDENTIARY LIMIT
- CONFIDENTIALITY → unauthorised disclosure
- INTEGRITY → unauthorised alteration / trust loss

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → identify the property before the remedy.

01 →

STAGE 01 CYBER-RISK CHAIN

CYBER-RISK CHAIN · THREAT ACTOR · EXPLOITS VULNERABILITY · SOCIAL CONSEQUENCE

THREAT ACTOR / EVENT → **EXPLOITS VULNERABILITY** → **OPERATIONAL / SOCIAL CONSEQUENCE** → **CAPABILITY → prevent / contain / recover / learn**

STARTING CONDITION / THREAT
- THREAT ACTOR / EVENT
- EXPLOITS VULNERABILITY

GOVERNANCE / RESPONSE MECHANISM
- OPERATIONAL / SOCIAL CONSEQUENCE
- CAPABILITY → prevent / contain / recover / learn

RIGHTS / STATUS / OUTCOME LIMIT
- THREAT ACTOR / EVENT
- EXPLOITS VULNERABILITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CAPABILITY → prevent / contain / recover / learn.

02 →

STAGE 02 INCIDENT-CRIME-OPERATION MATRIX

INCIDENT-CRIME-OPERATION MATRIX · CONTAIN AND RECOVER · I4C SUPPORT · INVESTIGATE AND PROSECUTE · STATE-LINKED STRATEGIC ACTIVITY · GRADED ATTRIBUTION · SECTION 66F CONDUCT + INTENT

THREAT / INSTITUTION / INSTRUMENT	DECISIVE LEGAL OR STATUS DISTINCTION	ATTRIBUTION / IMPLEMENTATION LIMIT
INCIDENT → CERT-In / entity → contain and recover	WARFARE → state-linked strategic activity → graded attribution	INCIDENT → CERT-In / entity → contain and recover
CRIME → police / I4C support → investigate and prosecute	TERRORISM → Section 66F conduct + intent	CRIME → police / I4C support → investigate and prosecute

THREAT / INSTITUTION / INSTRUMENT
- INCIDENT → CERT-In / entity → contain and recover
- CRIME → police / I4C support → investigate and prosecute

DECISIVE LEGAL OR STATUS DISTINCTION
- WARFARE → state-linked strategic activity → graded attribution
- TERRORISM → Section 66F conduct + intent

ATTRIBUTION / IMPLEMENTATION LIMIT
- INCIDENT → CERT-In / entity → contain and recover
- CRIME → police / I4C support → investigate and prosecute

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WARFARE → state-linked strategic activity → graded attribution.

03 →

STAGE 03 ATTRIBUTION CONFIDENCE LADDER

ATTRIBUTION CONFIDENCE LADDER · TECHNICAL INDICATOR · OPERATIONAL PATTERN · INFRASTRUCTURE CONTROL · LEGAL EVIDENCE · POLITICAL ATTRIBUTION · FALSE FLAGS AND COMPROMISED SYSTEMS REQUIRE QUALIFICATION

TECHNICAL INDICATOR

OPERATIONAL PATTERN / INFRASTRUCTURE CONTROL → **LEGAL EVIDENCE** → **POLITICAL ATTRIBUTION**

STARTING CONDITION / THREAT
- TECHNICAL INDICATOR
- OPERATIONAL PATTERN / INFRASTRUCTURE CONTROL

GOVERNANCE / RESPONSE MECHANISM
- LEGAL EVIDENCE
- POLITICAL ATTRIBUTION

RIGHTS / STATUS / OUTCOME LIMIT
- RULE → false flags and compromised systems require qualification
- TECHNICAL INDICATOR

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → false flags and compromised systems require qualification.

04 →

STAGE 04 IT ACT CII LADDER

IT ACT CII LADDER · SECTION 70 · PROTECTED SYSTEM · SECTION 70A · CII PROTECTION · SECTION 70B · INCIDENT RESPONSE · DESIGNATION AND INCIDENT HANDLING ARE NOT THE SAME FUNCTION

SECTION 70 → CII / PROTECTED SYSTEM

SECTION 70A → NCIIPC / CII PROTECTION → **SECTION 70B → CERT-In / INCIDENT RESPONSE** → **TRAP → designation and incident handling are not the same function**

STARTING CONDITION / THREAT
- SECTION 70 → CII / PROTECTED SYSTEM
- SECTION 70A → NCIIPC / CII PROTECTION

GOVERNANCE / RESPONSE MECHANISM
- SECTION 70B → CERT-In / INCIDENT RESPONSE
- TRAP → designation and incident handling are not the same function

RIGHTS / STATUS / OUTCOME LIMIT
- SECTION 70 → CII / PROTECTED SYSTEM
- SECTION 70A → NCIIPC / CII PROTECTION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → designation and incident handling are not the same function.

05 →

STAGE 05 CYBERCRIME GOVERNANCE CHAIN

CYBERCRIME GOVERNANCE CHAIN · ENTITY REPORT · POLICE REGISTRATION · I4C COORDINATION · CAPACITY SUPPORT · ELECTRONIC EVIDENCE

CITIZEN / ENTITY REPORT → **POLICE REGISTRATION / INVESTIGATION** → **I4C COORDINATION / CAPACITY SUPPORT** → **ELECTRONIC EVIDENCE → PROSECUTION / ADJUDICATION**

STARTING CONDITION / THREAT
- CITIZEN / ENTITY REPORT
- POLICE REGISTRATION / INVESTIGATION

GOVERNANCE / RESPONSE MECHANISM
- I4C COORDINATION / CAPACITY SUPPORT
- ELECTRONIC EVIDENCE → PROSECUTION / ADJUDICATION

RIGHTS / STATUS / OUTCOME LIMIT
- CITIZEN / ENTITY REPORT
- POLICE REGISTRATION / INVESTIGATION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: I4C COORDINATION / CAPACITY SUPPORT; ELECTRONIC EVIDENCE → PROSECUTION / ADJUDICATION.

06 →

STAGE 06 SECTORAL-RESILIENCE SYSTEM

SECTORAL-RESILIENCE SYSTEM · CII OWNER · ASSET INVENTORY · SECTOR REGULATOR · DOMAIN STANDARDS · CII COORDINATION · NATIONAL INCIDENT COORDINATION · ACCOUNTABILITY STAYS DISTRIBUTED BUT INTEROPERABLE

CII OWNER → asset inventory / continuity / controls

SECTOR REGULATOR / CERT → domain standards / response → **NCIIPC → CII coordination** → **CERT-In → national incident coordination**

THREAT / GOVERNANCE CONCEPT
- CII OWNER → asset inventory / continuity / controls
- SECTOR REGULATOR / CERT → domain standards / response

INSTITUTIONAL RESPONSE
- NCIIPC → CII coordination
- CERT-In → national incident coordination

LEGAL / EVIDENTIARY LIMIT
- RULE → accountability stays distributed but interoperable
- CII OWNER → asset inventory / continuity / controls

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SECTOR REGULATOR / CERT → domain standards / response.

07 →

STAGE 07 CERT-IN REPORTING BOUNDARY

CERT-IN REPORTING BOUNDARY · 28 APR 2022 · NOTICE OF SPECIFIED INCIDENT · REPORT WITHIN SIX HOURS · NOT PROVED · CRIME, CULPRIT OR FOREIGN ATTRIBUTION

28 APR 2022 → DIRECTIONS → **NOTICE OF SPECIFIED INCIDENT** → **REPORT WITHIN SIX HOURS** → **CONTAIN / COORDINATE / RECOVER** → **NOT PROVED → crime, culprit or foreign attribution**

THREAT / INSTITUTION / INSTRUMENT
- 28 APR 2022 → DIRECTIONS
- NOTICE OF SPECIFIED INCIDENT

DECISIVE LEGAL OR STATUS DISTINCTION
- REPORT WITHIN SIX HOURS
- CONTAIN / COORDINATE / RECOVER

ATTRIBUTION / IMPLEMENTATION LIMIT
- NOT PROVED → crime, culprit or foreign attribution
- 28 APR 2022 → DIRECTIONS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NOT PROVED → crime, culprit or foreign attribution.

08 →

STAGE 08 CYBER-LAW POWERS FIREWALL

CYBER-LAW POWERS FIREWALL · SECTION 69 · SECTION 69A · BLOCK PUBLIC ACCESS · SECTION 69B · COLLECT TRAFFIC DATA · SECTION 66A · STRUCK DOWN IN 2015

THREAT / INSTITUTION / INSTRUMENT	DECISIVE LEGAL OR STATUS DISTINCTION	ATTRIBUTION / IMPLEMENTATION LIMIT
SECTION 69 → intercept / monitor / decrypt	SECTION 69B → monitor / collect traffic data	RULE → purpose + procedure + proportionality
SECTION 69A → block public access	SECTION 66A → STRUCK DOWN IN 2015	SECTION 69 → intercept / monitor / decrypt

THREAT / INSTITUTION / INSTRUMENT
- SECTION 69 → intercept / monitor / decrypt
- SECTION 69A → block public access

DECISIVE LEGAL OR STATUS DISTINCTION
- SECTION 69B → monitor / collect traffic data
- SECTION 66A → STRUCK DOWN IN 2015

ATTRIBUTION / IMPLEMENTATION LIMIT
- RULE → purpose + procedure + proportionality
- SECTION 69 → intercept / monitor / decrypt

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SECTION 69B → monitor / collect traffic data.

09 →

STAGE 09 CYBERDOME MODEL

CYBERDOME MODEL · KERALA POLICE HUB · STATE COLLABORATION IS NOT CERT-IN OR NCIIPC · MUST REMEMBER · CYBER SECURITY SEPARATES ASSET AND CRITICAL-INFORMATION-INFRASTRUCTURE

KERALA POLICE HUB → **SPECIALISTS / INDUSTRY / ACADEMIA** → **RESEARCH / FORENSICS / AWARENESS / CAPACITY** → **LIMIT → State collaboration is not CERT-In or NCIIPC** → **MUST REMEMBER: Cyber security separates asset and critical-information-infrastructure**

THREAT / GOVERNANCE CONCEPT
- KERALA POLICE HUB
- SPECIALISTS / INDUSTRY / ACADEMIA

INSTITUTIONAL RESPONSE
- RESEARCH / FORENSICS / AWARENESS / CAPACITY
- LIMIT → State collaboration is not CERT-In or NCIIPC

LEGAL / EVIDENTIARY LIMIT
- MUST REMEMBER: Cyber security separates asset and critical-information-infrastructure...
- KERALA POLICE HUB

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Cyber security separates asset and critical-information-infrastructure.

10 →

STAGE 10 DPDP-VERSUS-CYBERSECURITY

PERSONAL DATA · FIDUCIARY OBLIGATIONS · IT ACT · CII PROTECTION · 13 NOV 2025 · RULES NOTIFIED; COMMENCEMENT STAGGERED · CLOSE DISTINCTION · CYBER INCIDENT IS NOT CYBERCRIME, VULNERABILITY IS NOT EXPLOIT

THREAT / INSTITUTION / INSTRUMENT	DECISIVE LEGAL OR STATUS DISTINCTION	ATTRIBUTION / IMPLEMENTATION LIMIT
DPDP → PERSONAL DATA / FIDUCIARY OBLIGATIONS / BOARD	NCIIPC → CII PROTECTION	CLOSE DISTINCTION: Cyber incident is not cybercrime, vulnerability is not exploit or...
IT ACT / CERT-In → INCIDENTS / OFFENCES / RESPONSE	13 NOV 2025 → RULES NOTIFIED; COMMENCEMENT STAGGERED	DPDP → PERSONAL DATA / FIDUCIARY OBLIGATIONS / BOARD

THREAT / INSTITUTION / INSTRUMENT
- DPDP → PERSONAL DATA / FIDUCIARY OBLIGATIONS / BOARD
- IT ACT / CERT-In → INCIDENTS / OFFENCES / RESPONSE

DECISIVE LEGAL OR STATUS DISTINCTION
- NCIIPC → CII PROTECTION
- 13 NOV 2025 → RULES NOTIFIED; COMMENCEMENT STAGGERED

ATTRIBUTION / IMPLEMENTATION LIMIT
- CLOSE DISTINCTION: Cyber incident is not cybercrime, vulnerability is not exploit or...
- DPDP → PERSONAL DATA / FIDUCIARY OBLIGATIONS / BOARD

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Cyber incident is not cybercrime, vulnerability is not exploit or.

11 →

STAGE 11 PYQ AND RESILIENCE RAIL

PYQ AND RESILIENCE RAIL · CYBERDOME UTILITY + LIMIT · CROSS-BORDER IMPACT + DEFENSIVE RESILIENCE · ELEMENTS + CHALLENGES + STRATEGY ASSESSMENT · EVIDENCE RUNG AND ATTRIBUTION CONFIDENCE · EVIDENCE LIMIT · NAME THE IT ACT OR NOTIFIED INSTRUMENT

2019 → CYBERDOME UTILITY + LIMIT → **2021 → CROSS-BORDER IMPACT + DEFENSIVE RESILIENCE** → **2022 → ELEMENTS + CHALLENGES + STRATEGY ASSESSMENT** → **END → IDENTIFY / PROTECT / DETECT / RESPOND / RECOVER / LEARN** → **QUALIFY → evidence rung and attribution confidence**

THREAT / GOVERNANCE CONCEPT
- 2019 → CYBERDOME UTILITY + LIMIT
- 2021 → CROSS-BORDER IMPACT + DEFENSIVE RESILIENCE

INSTITUTIONAL RESPONSE
- 2022 → ELEMENTS + CHALLENGES + STRATEGY ASSESSMENT
- END → IDENTIFY / PROTECT / DETECT / RESPOND / RECOVER / LEARN

LEGAL / EVIDENTIARY LIMIT
- QUALIFY → evidence rung and attribution confidence
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name the IT Act or notified instrument,...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name the IT Act or notified instrument,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name the IT Act or notified instrument,.

E →

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · NCIIPC (UNDER NTRO, SECTION 70A MANDATE) · IDENTIFIES CRITICAL · CERT-IN'S AUDIT AND EMPANELMENT ROLE · CERT-IN EMPANELS · SIX-HOUR REPORTING RULE · CERT-IN'S 28 APRIL 2022 DIRECTIONS · COORDINATES CYBERCRIME PREVENTION, CITIZEN REPORTING

OPTIONAL DOCTRINE / INSTITUTION DEPTH
- NCIIPC (under NTRO, Section 70A mandate): identifies critical
- CERT-In's audit and empanelment role: CERT-In empanels
- Six-hour reporting rule: CERT-In's 28 April 2022 Directions
- I4C: coordinates cybercrime prevention, citizen reporting.

LAW, ATTRIBUTION OR STATUS LIMIT
- Sectoral CERTs (CSIRT-Fin, CSIRT-Power): sector-specific incident
- Cyber Swachhta Kendra: the botnet-cleaning and malware-analysis
- CAUTION: Trusted-source telecom procurement: VisionIAS records that
- CAUTION: Telecommunications Act, 2023: replaced the Indian Telegraph Act,

COMPARATIVE GOVERNANCE / RIGHTS NUANCE
- DPDP Rules' phased architecture (2025-2027): immediate Data
- CAUTION: Current-anchor institutional detail: any specific audit count,
- CAUTION: The Huawei/ZTE supply-chain example as a live policy tension:
- CAUTION: Section 66A's post- Shreya Singhal status as a rights boundary

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels